

SAP BTP Migration Guide

The Future of PMM — Infrastructure Documentation

August 27, 2026 Chris O'Hara Target: SAP Business Technology Platform

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1. Executive Summary

The Future of PMM is an internal learning platform for SAP Product Marketing serving **~170 registered users**. The site includes a 12-unit curriculum, interactive assessments, AI-powered labs, and a weekly newsletter system.

Migration Goal: Move from Netlify + Supabase to SAP-managed infrastructure on Business Technology Platform (BTP), leveraging SAP Identity Authentication Service (IAS) for SSO.

2. Current Architecture Overview

2.1 Hosting & CDN

Component	Current Provider	Purpose
Static Hosting	Netlify	HTML/CSS/JS serving
CDN	Netlify Edge	Global content delivery
SSL Certificate	Netlify (auto-renewed)	HTTPS
Domain DNS	Network Solutions	DNS management

2.2 Backend Services

Service	Provider	Purpose
Database	Supabase (PostgreSQL)	User profiles, quiz scores, progress
Authentication	Supabase Auth	Magic link email login
Serverless Functions	Netlify Functions	API endpoints, AI integrations
Email Delivery	Resend	Newsletters, welcome emails

2.3 AI/External APIs

Service	Provider	Purpose
AI Text Generation	Anthropic Claude	Messaging Lab, BattleCoach
AI Text Generation	OpenAI GPT-4	GEO Lab, competitive analysis
Web Search	Brave Search API	Research tools, daily digest

3. Database Schema

3.1 Core Tables

```
-- User profiles (linked to Auth)
CREATE TABLE profiles (
  id UUID PRIMARY KEY,
  email TEXT UNIQUE NOT NULL,
  first_name TEXT,
  last_name TEXT,
  company TEXT DEFAULT 'SAP',
  role TEXT,
  team TEXT,
  newsletter_opt_in BOOLEAN DEFAULT true,
  created_at TIMESTAMPTZ DEFAULT now()
);

-- Quiz scores for curriculum units
CREATE TABLE quiz_scores (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES profiles(id),
  chapter TEXT NOT NULL, -- 'unit-01', 'unit-02', etc.
```

```
score INTEGER NOT NULL,
total INTEGER NOT NULL,
percentage INTEGER NOT NULL,
completed_at TIMESTAMPTZ DEFAULT now()
);

-- Assessment results (10x Scorecard, Cognitive Style)
CREATE TABLE assessment_results (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES profiles(id),
  assessment_type TEXT NOT NULL,
  result_key TEXT,
  result_name TEXT,
  result_data JSONB,
  completed_at TIMESTAMPTZ DEFAULT now()
);

-- Survey/feedback responses
CREATE TABLE survey_responses (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email TEXT,
  value_rating INTEGER,
  nps INTEGER,
  improvements TEXT,
  created_at TIMESTAMPTZ DEFAULT now()
);
```

3.2 Current Data Volume

Table	Records
Profiles	~170
Quiz Scores	~800
Assessment Results	~200
Survey Responses	~50

4. Serverless Functions (API Endpoints)

4.1 Authentication & User Management

Endpoint	Method	Purpose
/api/send-verification	POST	Send magic link email
/api/verify-token	POST	Validate magic link token
/api/subscriber-status	GET	Admin dashboard data

4.2 Progress Tracking

Endpoint	Method	Purpose
/api/save-result	POST	Save quiz/assessment results

/api/progress

GET/POST

Get/update user progress

4.3 AI Labs

Endpoint	Method	Purpose
/api/lab-ai	POST	Messaging Lab, BattleCoach (Claude)
/api/geo-scorecard	POST	GEO Visibility Lab (multi-LLM)

5. Third-Party Service Requirements

5.1 Service Comparison Matrix

Current Service	Purpose	SAP BTP Equivalent
Supabase	Database + Auth	SAP HANA Cloud + SAP IAS
Netlify	Static hosting	SAP Build Work Zone / Static Files
Netlify Functions	Serverless	SAP BTP Cloud Foundry / Kyma
Resend	Email	SAP internal SMTP / approved service
Anthropic Claude	AI	SAP AI Core (if available)
OpenAI	AI	SAP AI Core (if available)

5.2 Required API Keys

```
# Email Service
RESEND_API_KEY=re-xxxxx
FROM_EMAIL=chris@futureofpmm.com

# AI Services
ANTHROPIC_API_KEY=sk-ant-xxxxx
OPENAI_API_KEY=sk-xxxxx
BRAVE_API_KEY=BSAxxxxx

# Database
SUPABASE_URL=https://yyqzkczutlidhgyiyawc.supabase.co
SUPABASE_SERVICE_KEY=eyJxxxxx
```

6. Authentication Architecture

6.1 Current Flow (Magic Link)

1. User enters SAP email on login page
2. System sends magic link via Resend

3. User clicks link → JWT token generated
4. Token stored in localStorage (7-day expiry)
5. Token validated on each protected page

6.2 SAP Alternative: SAP Identity Authentication Service (IAS)

Benefits of SAP IAS:

- Single sign-on with existing SAP credentials
- Automatic user provisioning from SAP directory
- Automatic role/team mapping from SAP HR data
- Audit logging for compliance
- No separate password management

Note: Current system restricts to @sap.com email addresses only. SAP IAS would eliminate the need for separate account creation.

7. Static Assets & Content

7.1 File Structure

```
/dist/
├── index.html           # Homepage
├── curriculum/
│   ├── index.html      # Curriculum overview
│   ├── unit-01/ through unit-12/ # 12 unit pages
│   ├── labs/           # Interactive labs
│   └── signup.html      # Registration
├── newsletters/        # 29 newsletter issues
├── digests/            # 200+ daily AI digests
├── assessments/        # Interactive assessments
├── admin/              # Admin dashboard
├── assets/
│   ├── css/
│   ├── js/
│   └── images/
└── private/            # Gated content
```

7.2 Content Volume

Asset Type	Count
HTML Pages	~350 files
CSS/JS	~20 files
Images	~50 files

8. Migration Steps

- 1

Phase 1: Infrastructure Setup

Provision SAP HANA Cloud or PostgreSQL on BTP. Set up SAP IAS for authentication. Configure Cloud Foundry runtime for serverless functions.
- 2

Phase 2: Database Migration

Export data from Supabase (CSV/JSON). Create tables in SAP database. Import profiles, quiz scores, progress data. Verify data integrity.
- 3

Phase 3: Authentication Migration

Configure SAP IAS integration. Map existing users to SAP directory. Update frontend auth code. Test SSO flow.
- 4

Phase 4: Function Migration

Port Netlify Functions to SAP BTP runtime. Update environment variables. Configure AI service access (if approved). Update API endpoints.
- 5

Phase 5: DNS & Go-Live

Configure SAP CDN/hosting. Update DNS records. SSL certificate provisioning. Cutover with minimal downtime.

9. AI Service Considerations

9.1 Current AI Usage

Feature	Model	Monthly Volume
Messaging Lab	Claude 3.5 Sonnet	~500 requests
BattleCoach	Claude 3.5 Sonnet	~200 requests
GEO Scorecard	GPT-4 + Claude	~100 requests
Daily Digest	GPT-4 + Brave	~30 requests

9.2 Recommendation

The AI labs are valuable but **not critical** to core learning. Suggest:

- Core curriculum (Units 1-12) works without AI
- Labs marked as "AI-powered" with fallback content
- AI access added post-migration if/when approved

10. Environment Variables Summary

```
# Database
DATABASE_URL=<SAP_HANA_OR_POSTGRES_CONNECTION_STRING>
DB_SERVICE_KEY=<SERVICE_ACCOUNT_KEY>

# Authentication
SAP_IAS_CLIENT_ID=<IAS_CLIENT_ID>
SAP_IAS_CLIENT_SECRET=<IAS_CLIENT_SECRET>
SAP_IAS_TENANT=<TENANT_URL>

# Email (if using external service)
EMAIL_API_KEY=<APPROVED_EMAIL_SERVICE_KEY>
FROM_EMAIL=<APPROVED_SENDER_ADDRESS>

# AI Services (if approved)
ANTHROPIC_API_KEY=<IF_APPROVED>
OPENAI_API_KEY=<IF_APPROVED>
BRAVE_API_KEY=<IF_APPROVED>

# Admin
ADMIN_EMAILS=christopher.ohara@sap.com
```

Support & Contact

Site Owner	Chris O'Hara (christopher.ohara@sap.com)
Current Hosting	Netlify (Team: chrisohara99)
GitHub Repository	github.com/chrisohara99/futureofpmm
Supabase Project	yyqzkczutlidhgyiyawc

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